MILITARY SPECIFICATION

TOWER AB-585( )/G

1. SCOPE

1.1 This specification covers a guyed, lightweight, 106-foot, portable tower, designated Tower AB-585( )/G, (See 6.1 and 6.4.), composed of eleven 9.5-foot fold-up sections, and one half section, with all necessary tools and gear for installation. The complete tower with necessary erection equipment is transportable in one Truck, Cargo, 2½ Ton, 6 by 6, M35.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/3	Aluminum Alloy Bar, Rod, Shapes and Tube, Extruded, 2024
QQ-A-200/8	Aluminum Alloy Bar, Rod, Shapes and Tube, Extruded, 6061 and 6062
QQ-A-250/11	Aluminum Alloy 6061, Plate and Sheet
QQ-A-596	Aluminum Alloy Permanent and Semipermanent Mold Castings
QQ-A-601	Aluminum Alloy Sand Castings
WW-T-700/6	Tube, Aluminum Alloy, Drawn, Seamless, 6061 and 6062
QQ-S-763	Steel Bars, Shapes and Forgings, Corrosion Resistant
QQ-S-766	Steel Plates, Sheets, and Strip-Corrosion Resisting
QQ-S-781	Steel Strapping, Flat
GGG-H-86	Hammer, Hand, (Forged Steel Head)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-T-97	Tape, Pressure Sensitive Adhesive, Filament Reinforced

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-B-636	Box, Fiberboard
PPP-C-650	Crates, Wood, Open and Covered
PPP-S-760	Strapping, Nonmetallic (And Connectors)

MILITARY

MIL-C-104	Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-W-8604	Welding of Aluminum Alloys: Process for
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible
MIL-P-11268	Parts, Materials, and Processes Used In Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-22	Welded Joint Designs
MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MS 75014	Stake, Guy, Antenna (Stake GP-2)

DRAWINGS

ELECTRONICS COMMAND

SM-B-168754	Bracket
SM-B-168771	Tube

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

SM-B-168772	Tube
SM-C-423628	Guy Bag

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Tower AB-585( )/G is 4 feet by 4 feet by 106 feet high maximum and is guyed at four levels. Four additional guys are provided at the top to give high torsion and sway control. The tower is composed of fold-up sections fabricated from tubular aluminum. Eleven of the twelve sections have a nine-foot effective length. The top section is approximately one-half length. Extending diagonally across each full section is a sloping ladder for ease of climbing. A platform landing is located at the top of each section. The base of the tower consists of a frame with four adjustable feet for quick leveling.

3.2 Procurement model.- A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model, and the equipment shall incorporate all features of the model. (Note: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.3 Parts, materials, and processes; general.- In addition to the requirements of this specification, the requirements of Specification MIL-P-11268 including the selection requirements therein, shall apply.

3.3.1 Aluminum.- All vertical and davit tubing shall be drawn from aluminum alloy 6070-T6 as furnished by Aluminum Company of America, or equal. The tolerance on the outer diameter of the tubing shall be held to $\pm$.010 inch measured at any point. The wall thickness tolerance shall be held to $\pm$.006 inch measured at any point. All other tubular members shall be extruded or drawn aluminum alloy 6061-T6 or 6062-T6 conforming to WW-T-700/6 or QQ-A-200/8. All joint connections shall be fabricated by extrusion, forging, or stamping, but shall not be cast. All aluminum sheet and plate shall be 6061-T6 conforming to QQ-A-250/11. Castings where allowed, shall conform to QQ-A-601 or QQ-A-596.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

3.3.2 Steel.- All steel, unless otherwise specified herein, including nuts, bolts, pins, springs, bushings, etc., shall be type 301, 302, or 303, conforming to Specifications QQ-S-763 and QQ-S-766. Hardness and temper shall be as required to perform the intended function.

3.4 Finish, protective.- Equipment shall be given protective finish conforming to Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws etc., and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3). All exterior stainless steel shall be black oxide coated conforming to Specification MIL-C-13924.

3.5 Marking.- Marking shall conform to Specification MIL-M-13231.

3.5.1 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2 Installation caution notice.- Each tower shall be provided with an installation caution notice plate conforming to MIL-M-13231. Catagory (a), (b), or (c) material and process shall be used. The legend shall be marked on the plate with white or aluminum letters on a red background, as follows: "CAUTION - THIS EQUIPMENT SHALL NOT BE INSTALLED CLOSER THAN TWICE ITS HEIGHT FROM POWER LINES." The plate shall be attached to the base member of the tower so that it is readily visible from the ground as the equipment is installed.

3.6 Tower section (9 foot).- Each tower section shall be composed of three integral panels which fold up on vertical leg members to a maximum storage dimension of 9 feet 6 inches by 4 feet 2 inches by 5 inches. The fourth face shall be composed of four snap-on diagonal members constrained vertically on the legs by thrust rings. Each section shall include one snap-on platform and one snap-on ladder.

3.6.1 Three-panel assembly.- The three-panel assembly shall conform to the model. The panels shall be fabricated so that the 9 foot vertical tube length tolerance with respect to each leg length and positioning does not exceed $\pm$.025 inch. All snap-on diagonal member thrust rings shall be located to within $\pm$.025 inch from reference at the bottom of the nine-foot vertical tubes. The mounting centers of the vertical tubes on each face shall be 4 feet $\pm$.025 inch. The tubing size for the vertical members shall be 1.750 $\pm$.010 inches O.D. and wall thickness shall be .125 $\pm$.006 inch. The studs at the lower end of the vertical tubes shall conform to the model but the wall thickness shall be increased. The outside diameter of the parallel exposed portion of the stud shall be 1.435 $^{+.000}_{-.010}$ inches. The inside dimension shall be 1 1/8 $^{+.000}_{-.030}$ inches. Diagonal braces shall be 1.250 inches O.D. with .080 inch wall. The extruded fittings on the vertical tower legs that provide a smooth butt surface between panels during storage shall include a 3/32 inch radius on the outside to avoid interference during assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

3.6.2 Platform.- Each platform shall conform to the model. Frame locks shall fit smoothly over the horizontal section members. The frame lock mechanism shall be capable of rapid assembly without the use of a hammer or other tools. One platform shall include a trap door conforming to the model. Eleven platforms shall be included with each tower.

3.6.3 Ladder.- The section ladder shall conform to the model. Frame locks shall fit accurately and snugly in place and shall be capable of rapid assembly without the use of tools. Rung material shall be extruded aluminum 6062T6 conforming to Specification QQ-A-200/8. A lightweight hook device shall be provided to hold the ladder in the up position during erection so that the lower frame locks may be easily snapped down over the appropriate platform bar from the top side (four required for each tower).

3.6.4 Diagonal brace.- The diagonal brace shall conform to the model and shall meet the following requirements:

3.6.4.1 All diagonal braces shall be assembled from the inside of the tower.

3.6.4.2 The frame lock holes shall be held to 1.765 $^{+.005}_{-.000}$ inches. The pivot finger holes for fitting on the tower verticals shall be held to 1.765 $^{+.005}_{-.000}$ inches.

3.6.4.3 The hole in the pivot finger shall be located so that the 1 765 inch diameter on the pivot finger is coincident with the 1.765 inch diameter on the frame lock when in the operating position. Tolerances shall be controlled so that misalignment does not exceed $\pm$.005 inch. Over or under travel of the finger on the stop (when not in use) shall not exceed $\pm$.005 inch. The pivot finger shall be increased in the thickness to 7/16 inch minimum.

3.6.4.4 The tolerance for angularity of the frame locks to normal position shall not exceed $\pm 1\frac{1}{2}$ degrees in any plane. The vertical dimensions of the frame locks on the brace shall be held to a tolerance of $\pm$.015 inch. The horizontal dimensions of the frame locks on the brace shall be held to a tolerance of .025 inch. A frame lock bore parallelism requirement shall be within $\pm$.008 inch.

3.6.5 Section pin clamp.- Each tower section shall include four spring lock pins for connecting tower sections. The pin clamp shall conform to the model, but shall be scaled to function on a 1 3/4-inch diameter rather than 2-inch diameter tube. Holes for pin clamps in the tower leg shall line up properly when sections are fully engaged. Means shall be provided to confine the pin on the spring wire without stamping of the wire to prevent weakening of the spring.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
Check the source to verify that this is the currentversion before use.

3.7 Tower half section.- The special tower half section shall meet the requirements of 3.6.1 except the length of the panel verticals shall conform to the model. Diagonal members shall conform to 3.6.4. The special section shall be used only for capping the tower.

3.8 Top frame.- The top frame for terminating the top of the tower shall conform to the model. The studs shall have mounting hole center locations of 4 feet ±.025 inch for mounting to the tower section. Snap covers shall be furnished at the top of four vertical posts to prevent entrance of water.

3.9 Base assembly.- The base assembly shall be composed of four adjustable feet with legs and frame.

3.9.1 The feet shall conform to the model. The leg screw shall be aluminum alloy 2024-T4 conforming to Specification QQ-A-200/3. The foot shall properly function with an angularity of up to ±5 degrees from level.

3.9.2 The base frame shall conform to the model. Mounting center dimensions for mounting tower section studs shall be 4 feet ±.025 inch. The levels shall be mounted so that the bubbles indicate level when the four mountings holes are plumb. The carbon steel portions of the levels shall be replaced by a suitable noncorrosive material. Pin clamp storage holes shall be located horizontally rather than vertically so that the horizontal members of the first tower section, when assembled to the base, do not interfere with pin clamp removal. For protection the levels shall be mounted in a less vulnerable position. Collars shall be added at each end to provide protection against rough use.

3.10 Antenna mounting.-

3.10.1 Each tower shall include four antenna mounting post assemblies which shall conform to the model. The small end shall have dimensions specified on Drawing SM-B-168772 except the 5/8 inch in line holes shall be reduced to .385 inch diameter and shall be located 4.000 ±.005 inches from the shoulder which is located 7 inches from the end of the tube. The large end of the post shall conform to dimensions shown on Drawing SM-B-168771 except the 5/8-inch in line holes shall be reduced to .515 ±.005 inch and shall be located 3.500 inches from the shoulder that is located 6 inches from the end of the tube. The tolerance on hole location shall be ±.005 inch in both planes.

3.10.2 Eight stay bars, each including one 1 3/4 inch by 1 3/4 inch swivel coupler, shall be furnished and shall conform to the model. Each stay bar shall include a bracket assembly conforming to Drawing SM-B-168754.

3.10.3 Each tower shall include four "G" line attachments which shall conform to the model.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

3.10.4 Each tower shall include four "A" frame assemblies for mounting antennas. The "A" frame shall conform to the model with the following exceptions:

3.10.4.1 The connections for mounting the "A" frame to the tower leg shall permit four "A" frames to be mounted quickly and securely around the tower at any mounting level without interference of the mounting clamps and tightening handles.

3.10.4.2 The fold-up platform shall provide adequate horizontal bracing at the lower end of the "A" frame.

3.10.4.3 The flattened portions of the diagonal struts shall be lengthened at the ends and formed ½ inch around the tube prior to welding. A triangular gusset shall be welded on the inside of the strut on each end at the connection to provide adequate strength for rugged handling.

3.11 Light adaptor.- The corner post studs on the top frame (3.8) shall be the light adaptor mounting and shall conform to the model.

3.12 Auxiliary horizontal members.- Eight auxiliary horizontal members shall be furnished with each tower. The horizontal member shall include two 1 3/4 inch by 1 3/4 inch swivel connections which shall conform to the model.

3.13 Davit assembly.- The davit assembly shall conform to the model with the following exceptions:

(a) The davit spring lock pin for securing the two three inch O.D. tubes shall be constructed so that the lock pin holds the sections securely together after repeated use. The pin shall be type 302 steel conforming to Specification QQ-S-763. The lock pin assembly shall be designed so that the pin is replaceable for maintenance purposes.

(b) The davit tube size shall be 3 inches O.D. by .125 inch wall.

(c) The sheaves for the hauling rope shall be 3 inches O.D. minimum and shall include suitable oilite-bronze bushings to minimize friction during erection.

(d) The thrust bearing at the base of the davit shall be designed so that adequate clearance to the tower is provided when the davit is moved from section to section of the tower during functional use.

(e) The thrust clamp shall function with the davit mounted on any vertical leg of the tower.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
Check the source to verify that this is the currentversion before use.

(f) The stud diameter on the lower three inch tube shall permit a good fit with the 2.750 inch I.D. tube. The end of the stud shall include 3/32 inch by 45 degree chamfer to provide ease of entry.

(g) A discharge device shall be provided to eliminate static-electricity charge during erection, when the tower sections are being hoisted by the davit. Two 1/8-inch, 7 by 19, stainless steel aircraft cables shall be suitably mounted to the davit arm and suspended three feet. Spacing shall be arranged to provide most effective use of the discharge device.

3.13.1 Hauling rope.- The 250-foot hauling rope shall conform to the model except that the rope shall be 3/8 $^{+.010}_{-.015}$ inch diameter, olive drab, three strand, yacht lay, prestretched, stabilized filament dacron (polyester fibre); minimum breaking strength 2800 pounds; maximum allowable elongation 15 percent at 840 pounds, as made by Columbian Rope Company, Auburn, N.Y., or equal.

3.13.2 Davit guide clamp.- The davit assembly shall include two davit guide clamps for mounting and moving the davit along the vertical leg of the tower. The davit clamp shall conform to the model except as follows:

(a) The cam lock shall include positive means to provide adjustment for normal wear of the cam.

(b) The two tubes for connecting the davit guide bearings shall be increased in wall thickness by at least .020 inch to provide a more rugged assembly for rough handling.

(c) The davit hauling rope shall conform to the model with the exceptions specified in 3.13.1. The hook shall conform to the model.

(d) The davit guide bearings shall be of such dimensions that when the guide is clamped to the tower vertical leg, the three inch davit tube shall move freely and smoothly in rotation or axial movement.

3.13.3 Sling.- The sling shall conform to the model except that the rope shall be $\frac{1}{4}$ $^{+.010}_{-.015}$ inch diameter, olive drab, three strand, yacht lay, prestretched, stabilized filament dacron (polyester fibre); minimum breaking strength 1300 pounds; maximum allowable elongation 15 percent at 390 pounds, as made by Columbian Rope Company, Auburn, N.Y., or equal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

3.14 Guy assembly.- All guy assemblies shall conform to the model. Guy material shall be 3/16 inch 7 by 19 aircraft cable with Zytel 37 X Nylon jacketing $\frac{1}{4}$ inch diameter, (rated) breaking strength 4200 pounds minimum, as made by the Rochester Rope Co., Culpepper, Va., or equal. All guy wire terminations shall be assembled in such a manner that failure shall not occur within the rated strength of the cable. Guys shall be color coded conforming to the model and the length of the guy shall be stamped on each guy. (See 4.8).

3.14.1 Guy quantities shall be as follows:

(a) 8 ea. 162 feet (Red)
(b) 4 ea. 136 feet (White)
(c) 4 ea. 78 feet (Blue)
(d) 4 ea. 60 feet (Green)

3.14.2 Tension indicator.- Each guy shall include a tension indicator conforming to the model except as follows:

3.14.2.1 The length of the attachment loops shall be increased to allow easy entrance of the anchor attachment hook and guy hook.

3.14.2.2 The indicator spring shall be preloaded to 300 pounds.

3.14.2.3 Calibrations shall be set at zero and 180 degrees about the periphery to allow good readability. Factory adjustment shall be positively locked to preclude loss of adjustment. Marking shall conform to Figure 1.

3.14.2.4 Calibrations shall be accurate to ± 8 percent. (See 4.11)

3.14.2.5 The device shall "bottom out" at 500 pounds with the spring not fully compressed and within the elastic limit of the spring material.

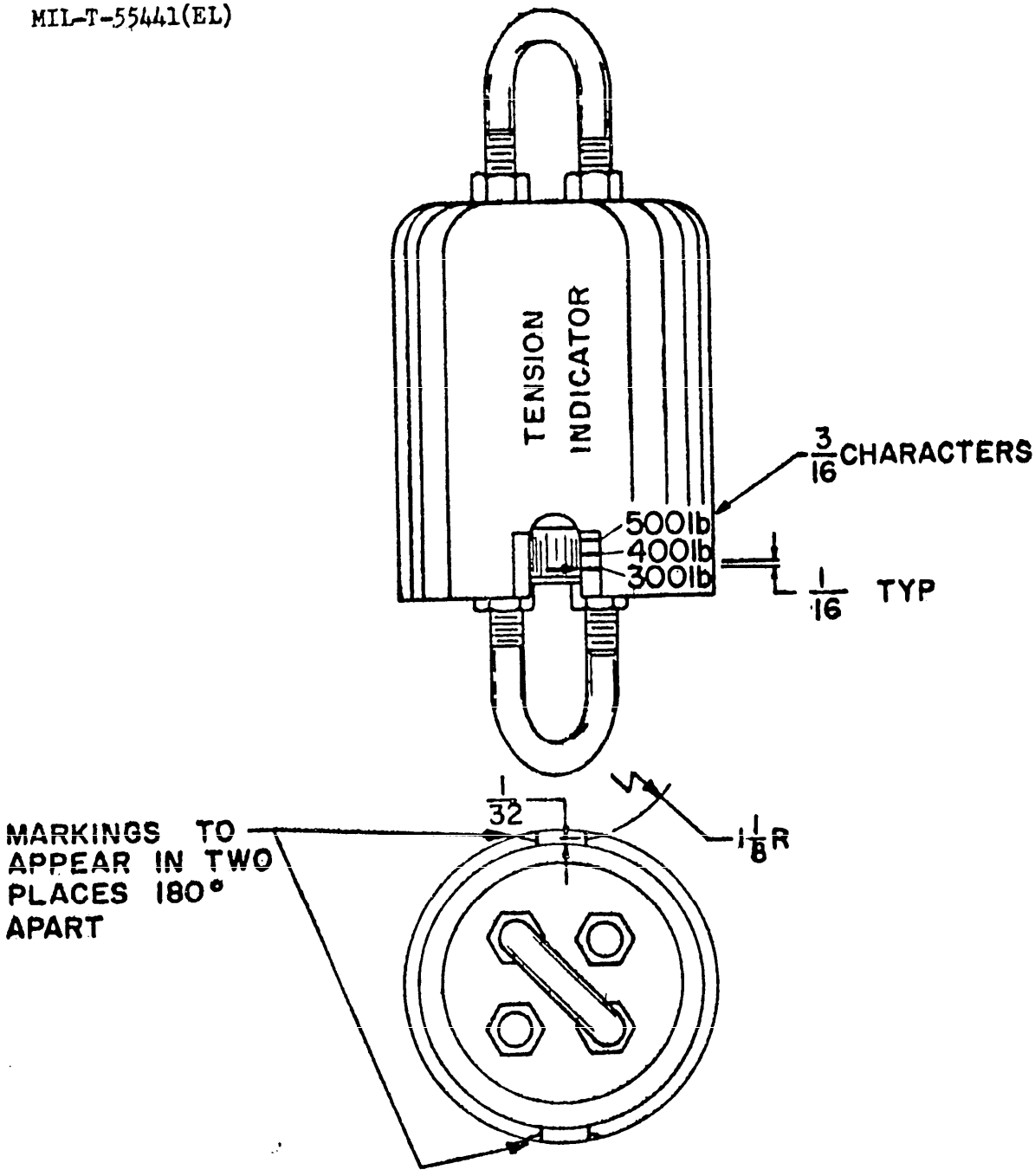
3.14.2.6 Calibration lines shall be placed on a 45 degree chamfered plane so that the scale lines and the indicator may be more easily read.

3.14.3 Guy tensioning device.- The guy tensioning device shall conform to the model. The engravings on the nut shall be filled with contrasting color to improve readability.

3.14.3.1 Spanner tool.- Eight spanner tools shall be furnished with each Tower AB-585( )/G and shall conform to the model. The tool is for final tightening of the guy tensioning device (3.14.3).

3.15 Anchoring.- Each tower shall be supplied with four oval eye screw anchors, No. 7528-A, 8-inch, and four No. 7530-A, 10-inch, as made by Hubbard and Co., Pittsburg, Pa., or equal.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.



TENSION INDICATOR MARKINGS
FIG. 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

3.15.1 Each anchor shall include attachment clevis and guy attachment assemblies conforming to the model. The clevis shall be type No. J-705 as made by Joslyn Co., New York City, or equal.

3.15.2 Stakes.- Each tower shall include 16 each Stake GP-2 conforming to Standard MS 75014.

3.15.3 Digging bar.- The digging bar shall conform to the model. Two digging bars shall be furnished with each Tower AB-585( )/G.

3.16 Transit mounting.- 2 inch wide webbing straps with necessary D rings, hooks and buckles shall be supplied for mounting Tower AB-585( )/G in Truck, Cargo, 2½ Ton, 6 x 6, M35. Strapping shall be color coded for ease of use. One bag per tower shall be supplied to contain the davit, couplers, snatch bolck and slings during transportation, and one bag shall be supplied to contain sixteen stakes GP-2. The bags shall be fabricated from cotton duck and include all necessary straps, buckles and draw strings.

3.17 Cable clamps.- Thirty-six r-f cable clamps shall be furnished with each tower. The clamps shall conform to the model.

3.18 Tag line.- The tag line shall conform to the model with the exceptions specified in 3.13.1.

3.19 Anchor location rope.- The 80-foot anchor-location rope shall conform to the model with the exceptions specified in 3.13.3.

3.20 Hammer.- Each tower shall include one hammer, Type X blacksmiths, Class 1, double-face, 6 pound, conforming to Federal Specification GGG-H-86.

3.21 Miscellaneous equipment carrier.- An accessories bag in accordance with Drawing SM-C-423628 shall be provided for carrying miscellaneous equipment such as cable clamps, anchor-location rope, ground wire, etc.

3.22 Tower grounding.- Tower grounding connections shall be furnished on each section conforming to the model. Four 5/8 inch diameter by 8-foot long copperweld ground rods with wire and hardware shall be furnished conforming to the model. Grounding fittings that are mounted on each section shall not interfere during assembly. The wing nuts shall be made captive to prevent loss.

3.23 Weight.- One complete tower section including platform, ladder, and diagonal braces shall not exceed 126 pounds. The breakdown shall be as follows:

(a) Platform 31.5 pounds maximum

(b) Ladder 13.5 pounds maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

(c) Three panel assembly 72 pounds maximum

(d) Diagonal brace (four) 9.5 pounds maximum

3.24 Drainage.- All tower members shall be designed to prevent water from becoming trapped and bursting the tubes during frigid weather when the tower is in use or in the stored condition.

3.25 Loading.- The tower shall be fabricated to withstand 100 mph wind velocities at full height with four six-foot paraboloidal or two eight-foot paraboloidal antennas mounted about the top of the tower with all "A" frames mounted on the special top half section. (See 4.6).

3.25.1 Davit loading.- The davit shall be capable of withstanding a load of 750 pounds in the fully extended position. The davit in the heavy duty position (bottom section removed) shall be able to sustain a load of 1000 pounds. (See 4.7).

3.25.2 Diagonal brace loading.- Each snap-on diagonal brace (3.6.4) shall, in its normal assembled position, be capable of withstanding a direct horizontal pull of 600 pounds applied 3 inches from the tower leg toward the inside of the tower and in line with the opening of the frame lock. The brace shall also be capable of withstanding an impact of 75 foot pounds caused by a falling weight. The impact shall be applied from the outside of the tower toward the snap-on face and on an area within 6 inches of the vertical leg. (See 4.9).

3.25.3 Platform loading.- The platform grating shall be capable of withstanding a concentrated load of 800 pounds on 25 square inches of surface at any point. (See 4.10).

3.26 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.13). Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.27 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.28 Preproduction samples.- The contractor shall furnish one preproduction sample for approval, if required by the invitation for bids and contract (See 4.3 and 6.2 (e) (1).)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

3.29 Welding.- Wherever practicable, welded joints shall conform to MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. All welds shall conform to Specification MIL-W-8604. (See 4.12).

3.30 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268

Riveting.
General requirements for securing of parts.

(b) In this specification

(3.4) Finish.
(3.5) Marking.
(3.29) Welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (see 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.15.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the currentversion before use.

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B (see tables II and III respectively). Preproduction inspection shall be performed in any order which is satisfactory to the Government.

Table I - Preproduction Inspection

Inspection (For additional preproduction inspection See 4.3)	Rqt. Para.	Insp. Para.
Tower Load	3.25	4.6

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4, 4.5.1 and 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government-at its discretion——may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

Table II - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.30	4.14	1.0%	4.0%

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 Group B sampling plans.- The group B sampling plans, for the AQL's listed in Table III shall be as follows:

AQL	Special Inspection Plan
10.0%	S-3

4.5.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Guy	3.14	4.8	10.0%
Tension indicator accuracy	3.14.2	4.11	10.0%
Davit	3.25.1	4.7	10.0%
Diagonal brace	3.25.2	4.9.1 and 4.9.2	10.0%
Platform load	3.25.3	4.10	10.0%
Interchangeability	3.26	4.13	10.0%
Welding	3.29	4.12	10.0%

4.5.3 Reinspection of conforming group B sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B inspection may be accepted on contract or order, provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

4.6 Tower load test.- Three sections of the tower shall be assembled vertically on the tower base. The special 4.5 foot section with two "A" frames shall be mounted on top of the three sections. Four suitable anchors shall be located 24 feet from the centerline of the tower on level ground. The anchors shall be located on lines at 45 degrees to the perpendicular to each face of the tower and shall be spaced at 90 degrees. Eight test guys shall be attached to the lower guy plate location of the top 4.5 foot section of the tower. A pair of guys shall be located diagonally opposite on the guy plates of each tower corner and the four pairs of guys shall converge to each of the four anchors of the tower. The guys shall be initially loaded to about 400 pounds each. A simulated wind load of 175 pounds for each five feet of tower shall be imposed along the tower normal to one face. A compression force of 12,000 pounds shall be then applied directly to the top of the tower by means other than the eight guys. In addition, a 2200-pound load shall be applied in the direction of the simulated wind load to the vertical post, at the top, to the "A" frame. An additional 2200 pounds shall be applied also in the direction of the simulated wind load but to the top of the "A" frame post on the opposite face of the tower. The tower shall be oriented in this test such that the snap-on members are subjected to the worst load conditions. No failure or permanent deformation shall occur with this loading. (See 3.25).

4.7 Davit tests.- The davit shall be statically loaded in accordance with 3.25.1. During this test, the davit shall be mounted to the tower in the same manner as in actual use. No permanent deformation shall occur to the tower members or to the davit assembly.

4.8 Guy test.- The guys shall be tested to the rated loading of 3.14. No guy hardware shall fail or slip prior to the rated strength of the cable. The guy adjustment mechanism shall not damage the cable or cable jacket after 72 hours at the rated cable load. The tension shall be checked and adjusted at 24 hour intervals.

4.9 Diagonal brace test.- (See 3.25.2).

4.9.1 Impact.- A 25-pound weight shall be dropped from a height of three feet on the snap-on diagonal brace approximately six inches from the center line of the tower leg. The snap-on brace shall be mounted in such a manner that the force is directed from the outside of the tower and normal to the tower face. No failure or disengagement shall occur.

4.9.2 Static.- A 600-pound force shall be applied to the snap-on diagonal brace six inches from the center line of the tower leg. The force shall be directed toward the inside of the tower and normal to the face. No failure or disengagement shall occur.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.10 Platform test.- The platform, when mounted on the tower section, shall be tested and proven capable of supporting a load of 800 pounds on 25 square inches of surface at any point without permanent deformation. (See 3.25.3).

4.11 Tension indicator.- The tension indicator shall be tested and proven to be accurate to ±8 percent of the true load when tested respectively at 300 pounds, 400 pounds and 500 pounds. (See 3.14.2).

4.12 Welding inspection.- Welding shall be inspected in accordance with Specification MIL-W-8604. (See 3.29).

4.13 Inspection for dimensional interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.26. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) Tower section overall fit.

(b) Individual tower leg stud fits.

(c) All frame lock to tube fits.

(d) Diagonal brace to tower fit.

(e) Platform to tower section fit.

(f) Platform ladder mounting tube location fit.

(g) Ladder to tower section fit.

(h) Top and base frame to tower section fit.

(i) Davit guide bearing and thrust bearing fits to davit boom.

(j) Davit stud fit and hole locations.

(k) Trap door to platform fit.

(l) "A" frame fits to tower.

(m) Antenna mounting post fits and stud dimensions.

(n) Tower interlock pin fits.

(o) Interchangeability of all replaceable items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

4.14 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table IV.

Table IV - Classification of Visual and Mechanical Defects

Classification	Defect
Major	Loose nuts and screws. Welding, such as scabs, blisters, pock marks and other harmful defects. Riveting, rivet too close to edge of material riveted. Painting, color not as specified, unsatisfactory adhesion, film thickness less than ½ specified thickness. Marking missing, illegible or incorrect. Parts, missing or improperly assembled. Item not within requirements of drawing.
Minor	Scratches, cuts, abraisions peeling or chipping. Useless holes or slots.

4.15 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance with the applicable provisions of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified.

5.1.1 Level A.-

5.1.1.1 Cleaning.- Cleaning shall be accomplished in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Drying shall be accomplished in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Unit Packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

5.1.1.3.1 Technical literature.- Each technical literature shall be packaged method 1C-1.

5.1.1.3.2 Running spares, accessories and ancillary items.- Running spare parts, accessories and ancillary items as required, shall be packaged in accordance with MIL-E-17555.

5.1.1.3.2.1 Consolidated package.- Consolidate the items packaged as specified above within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Fill all voids with paperboard conforming to PPP-P-291, type III to prevent movement. Box closure shall be as specified in the appendix of the box specification.

5.1.1.3.3 Components of Tower AB-585( )/G.- Packaging shall be developed in accordance with the methods listed below, employing kinds and types of materials which when applied will afford adequate protection against corrosion, deterioration and damage during world-wide shipment, handling and open storage:

METHOD III

All cushioning shall be fabricated of material equal to the weather-resistant requirements of the unit package container. The unit container shall conform to the requirements of PPP-B-636, type CF, class weather-resistant. Closure shall be in accordance with the appendix of the box specification. In addition, all seams and joints shall be sealed with tape, not less than 2 inches wide, conforming to PPP-T-76.

5.1.2 Level C.- Each component of Tower AB-585( )/G shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be level A, B or C as specified.

5.2.1 Level A.- Components of Tower AB-585( )/G packaged as specified in 5.1, shall be packaged in containers selected from Table V, based upon the most economical container required to provide minimum, adequate protection for the packaged item for this packing level of protection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

TABLE V

CONTAINER REQUIREMENTS

Container	Specification	For Level A	For Level B	Max. Gross Wt, Limitation
Crates, Wood, Open and Covered	PPP-C-650	Style B, Type V Covered	Style B, Type V Covered	Table V (Note B

NOTE B.- Covered crates conforming to PPP-C-650 shall be used only when the weight of a single item exceeds 1000 pounds. Sheathed crates conforming to MIL-C-104 shall be used only when the weight of a single item exceeds 4000 pounds.

5.2.1.1 Container waterproofing.- When the containers being placed directly into the shipping container are not waterproofed, waterproofing shall be effected by use of an individual wrap or a caseliner conforming to MIL-L-10547, grade B, and appendix thereto, prior to packing. Dunnage not inclosed within the waterproof wrap or caseliner shall consist of materials treated to resist water. Wraps or caseliners may be omitted from fiberboard boxes provided all seams, corners and manufacturer's joint are waterproofed by sealing with tape, not less than 2 inches wide, conforming to PPP-T-76. The tape shall be centered over the seams and joints and shall extend over all corners and edges of the box a minimum of two inches onto the adjacent box panels. Tape shall be applied over the lengthwise seams of the outer flaps closing the openings of the box and over the manufacturer's joint prior to tape being applied to the edge seams of the box. The tape applied to the manufacturer's joint shall only fully cover the joint and not be extended over the corners of the box onto the adjacent panels.

5.2.1.2 Container closure.- Closure shall be in accordance with the applicable container specification or appendix thereto.

5.2.1.3 Strapping and reinforcing.- Metal strapping conforming to QQ-S-781, type I, class B shall be applied to wood and wood-cleated shipping containers in accordance with the requirements of the applicable box specification or appendix thereto. Fiberboard shipping containers shall be reinforced by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II respectively; selection of the material and applicati shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B.- Components of Tower AB-585( )/G, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that Level B containers of Table V shall apply and 5.2.1.1 shall not apply.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

5.2.3 Level C.- Components of Tower AB-585( )/G shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Inspection.- Inspection of military packaging shall be in accordance with MIL-P-116.

6. NOTES

6.1 Intended use.- Tower AB-585( )/G is intended to be used for supporting microwave and other type antennas for use with radio relay sets and other communication equipment where height, stability, multiplicity of antenna mounting and fast erection are required. The tower is intended to be used also at semi-permanent installations when extra height and rigidity over standard masts is required.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specificati on and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) One sample unit of each item cited in section 1 is generally required. (See 3.28)

(2) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

(g) Place of final inspection.

(h) Technical literature required (See 3.27).

(i) Quantity of tools and running spare parts required. (See 3.27).

(j) Quantity production. When the procurement quantity for this tower exceeds 100, the method of fabrication of the diagonal brace connector fittings on diagonal braces on all three panel assemblies shall be as follows:

(1) The formed stamping or forging technique shall be used to fabricate the fitting rather than the trimmed extrusion technique.

(2) The technique shall be such that the connector between the diagonal tube and the fitting is a tube within a tube rather than two trimmed extruded strips in a squared hole as is provided on the model.

6.3 Color.- The color chip furnished by the procuring agency (See 3.4) will be color chip No. X-24087, and may be obtained upon request to Commanding Officer, Electronics Support Command, ATTN: SELSC/PO-TD, 225 South Eighteenth St., Philadelphia, Pennsylvania 19103.

6.4 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AB-585W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.5 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army-EL

Preparing Activity:
Army-EL

Project No. 5445-A001

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

ESC-FM 2011-66

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:17Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
HEADQUARTERS
U.S.ARMY ELECTRONICS COMMAND
FORT MONMOUTH, NEW JERSEY 07703

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARM

Commanding General
U. S. Army Electronics Command
ATTN: AMSEL-PP-
Fort Monmouth, New Jersey 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:17Z
Check the source to verify that this is the current version before use.